

Cross-contamination versus undercooking of poultry meat or eggs - which risks need to be managed first?

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Epidemiological studies show that poultry meat and eggs are important sources for consumers' exposure to pathogens such as Salmonella and Campylobacter. There is a focus in many countries to reduce the level of human illness from food-borne pathogens. Reduction of the prevalence of contaminated poultry meat or eggs is one major area of focus. The other is risk communication to the consumer, where information aimed at changing the food preparation behaviour has been utilised as a risk management tool. The efficacy of messages such as 'cook poultry meat and eggs thoroughly' or 'wash your hands' will depend both on the ability to change consumer behaviour as well as where the risk can best be mitigated. In order to prioritise what message should be given to the consumer, the relative contribution of different exposure pathways finally leading to ingestion of the pathogens and resulting in illness needs to be known. It is important to know whether cross-contamination events or undercooking are the greatest risk lurking in consumers' kitchens. A review of studies looking at the location of pathogens in food products has been performed and data regarding internal and external (surface) contamination of poultry meat with Salmonella spp. and Campylobacter jejuni and C. coli is presented. In the case of eggs, data on internal contamination with Salmonella and for contamination of egg shells with Salmonella and Campylobacter are discussed. The results from published risk assessments for these pathogen-food commodity combinations have been evaluated and conclusions regarding the relative risk of internal and external contamination of poultry meat and eggs were drawn. In conclusion, cross-contamination events from activities such as use of the same cutting board for chicken meat and salad without intermediate cleaning or spreading of pathogens via the kitchen environment seem to be of greater importance than the risk associated with undercooking of poultry meat or eggs. Risk management options are discussed against the background of risk communication strategies used in different countries.